

electron/main.cjs

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Overview

A new `electron/main.cjs` file replaces the previous main-process entry point. It imports Node modules, sets up the Electron app, defines window helpers, and registers a comprehensive IPC surface.

Key changes

- **Imports** added at lines 1-16: `fs/promises`, `fs`, `path`, `url`, `electron`, `compare-runner.cjs`, `llm.cjs`, `prefetch-summaries.cjs`, etc. (R1-16).
- **Window helpers:**
 - `createWindow` (lines 164-203) creates the main window.
 - `buildMenu` (144-162) builds the application menu.
 - `iconPath` (136-142) resolves the app icon.
 - `windowStatePayload` (118-123) and `emitWindowState` (129-133) report window state.
- **Progress emitters:**
 - `sendEngineProgress` (18-22)
 - `sendKnowledgeGraphProgress` (571-587)
 - `sendWorkspaceHistoryProgress` (806-812)
 - `sendGithubAuthProgress` (838-844)
- **IPC surface:** hundreds of `ipcMain.handle` registrations covering:
 - Compare operations (`compare-folders`, `get-file-diff`, `filter-changes-gitignore`, etc.) – lines 222-457.
 - LLM actions (`llm-summarize`, `llm-summarize-stream`, `llm-abort-stream`, `llm-probe`) – lines 306-335.
 - Knowledge-graph build/read – lines 589-611.
 - Docs handling (`write-novadiff-docs`, `read-novadiff-docs-file`, `open-novadiff-docs-in-browser`) – lines 500-570.
 - Git/GitHub integration (`git-detect-tooling`, `git-repo-status`, `github-list-repos`, `github-pr-compare-roots`, `github-gh-install`, etc.) – lines 694-852.
 - Workspace management (`workspace-create`, `workspace-set-active`, `workspace-list`, `workspace-update-live-repo`, `workspace-ensure-commit-snapshot`, etc.) – lines 898-1069.
 - Snapshot operations (`workspace-snapshot-list-files`, `workspace-snapshot-read-file`) – lines 1027-1056.
- **Lifecycle hooks:** `app.whenReady` (205-208), `activate` (209-213), `window-all-closed` (216-220), and `before-quit` (1088-1090) which calls `cleanupNovadiffWorktrees`.

Impact

- IPC channel names are explicit; duplicate names would override earlier handlers (risk: collision).
- Path validation in `read-novadiff-docs-file` (521-549) and `knowledge-graph-read-file` (646-660) mitigates traversal attacks.
- Git clone and GH install use `child_process.spawn` with sanitized arguments; injection remains a risk if inputs are not validated.
- Node 18
- `node:` imports require compatible Electron; older builds may fail.
- Progress emitters provide telemetry but need matching listeners to avoid memory leaks.

Risks & follow-ups

1. Verify renderer code does not reuse IPC channel names such as `compare-folders` or `llm-summarize`.
2. Test edge cases for path validation in docs and graph read handlers.
3. Ensure inputs to `spawn` (git clone, gh install) are sanitized to prevent command injection.
4. Confirm that `nova-diff-icon.png` is bundled for packaged builds.